Student Performance Prediction Using Machine Learning

Abstract

The objective of this project is to predict student mathematics performance using demographic and academic factors. Machine learning techniques were applied to analyze the relationship between various student attributes and mathematics scores. The project involved data preprocessing, exploratory data analysis, model development, and evaluation. A Linear Regression model was trained and achieved an R^2 score of 88.4%, demonstrating strong predictive performance.

Introduction

Educational institutions often seek ways to identify factors that influence student performance. Predictive analytics can help educators recognize patterns and provide timely support to students. This project focuses on predicting mathematics scores based on student demographic information, parental education level, lunch type, test preparation status, and other academic indicators.

Dataset Description

The dataset contains information about students, including:

- Gender
- Race/Ethnicity
- Parental Level of Education
- Lunch Type
- Test Preparation Course
- Mathematics Score
- Reading Score
- Writing Score

The dataset consists of 1000 student records and multiple features that influence academic performance.

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
0	female	group B	bachelor's degree	standard	none	72	72	74
1	female	group C	some college	standard	completed	69	90	88
2	female	group B	master's degree	standard	none	90	95	93
3	male	group A	associate's degree	free/reduced	none	47	57	44
4	male	group C	some college	standard	none	76	78	75

Methodology

Data Collection

The Students Performance Dataset was used for this project.

Data Preprocessing

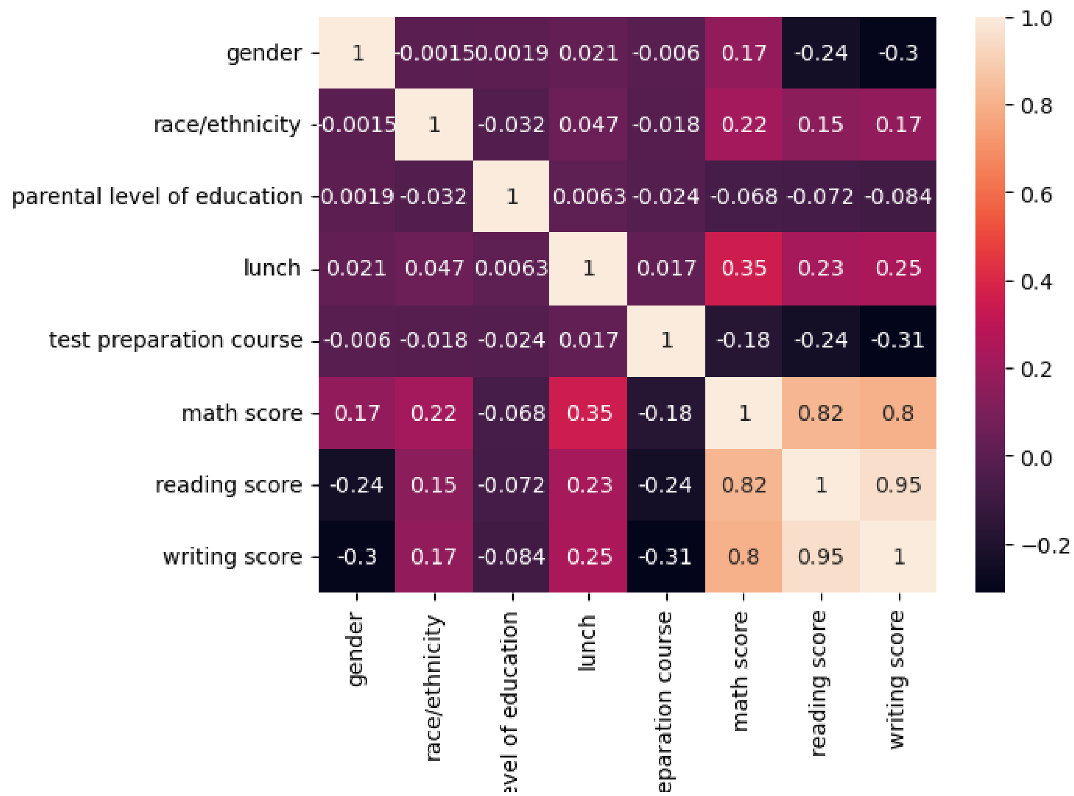
The following preprocessing steps were performed:

- Checked for missing values
- Verified data consistency
- Encoded categorical variables using Label Encoding
- Prepared features and target variables

Exploratory Data Analysis

Several visualizations were created to understand the dataset:

- Distribution of mathematics scores
- Boxplots for performance comparison
- Correlation heatmap



Model Development

A Linear Regression model was selected to predict mathematics scores.

Training steps:

1. Feature selection
2. Train-test split (80:20)
3. Model training
4. Prediction generation

Evaluation Metrics

The model was evaluated using:

- Mean Squared Error (MSE)
- R^2 Score

Results

The Linear Regression model produced the following results:

- Mean Squared Error (MSE): 28.28
- R^2 Score: 0.884

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MSE: 28.275284506327342
R2 Score:  0.8838026201112223
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These results indicate that the model successfully explains approximately 88.4% of the variation in mathematics scores.

Key Insights

1. Reading scores have a strong positive relationship with mathematics performance.
2. Writing scores significantly influence mathematics achievement.
3. Students who completed test preparation courses generally perform better.
4. Parental education level positively affects academic outcomes.
5. Students receiving standard lunch tend to perform slightly better than those receiving free or reduced lunch.

Recommendations

1. Encourage participation in test preparation programs.
2. Improve reading and writing skill development initiatives.

3. Identify low-performing students early for targeted support.
4. Increase parental involvement in educational activities.
5. Provide additional assistance to economically disadvantaged students.

Conclusion

This project successfully demonstrated the use of machine learning for predicting student mathematics performance. The Linear Regression model achieved strong predictive accuracy with an R^2 score of 88.4%. The findings highlight the importance of reading ability, writing skills, parental education, and test preparation in influencing student success. These insights can assist educators and institutions in making data-driven decisions to improve academic outcomes.

Future Scope

- Apply advanced machine learning algorithms such as Random Forest and XGBoost.
- Deploy the model as a web application.
- Use larger and more diverse educational datasets.
- Explore additional factors affecting student performance.